

Chapter 4: Quantum teleportation

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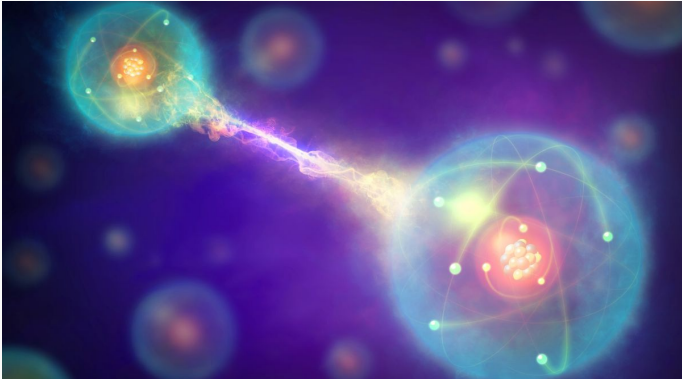
Quantum Teleportation

Quantum Entanglement : A review

"When two or more particles link up in a certain way, no matter how far apart they are in space, their states remain linked.

That means they share a common, unified quantum state.

So observations of one of the particles can automatically provide information about the other entangled particles, regardless of the distance between them." - Paul Sutter, 'What is quantum entanglement?'



Quantum Entanglement : A review

Definition

A state $\rho \in D(\mathbb{H}^A \otimes \mathbb{H}^B)$ in a composite system $\mathbb{H}^A \otimes \mathbb{H}^B$, which is composed of the sub-systems $\mathbb{H}^A$ and $\mathbb{H}^B$ is called **separable** with respect to the sub-systems $\mathbb{H}^A$ and $\mathbb{H}^B$, if there exist states $\rho_j^A \in D(\mathbb{H}^A)$ in sub-system $\mathbb{H}^A$ and $\rho_j^B \in D(\mathbb{H}^B)$ in sub-system $\mathbb{H}^B$ indexed by $j \in I \subset \mathbb{N}$ together with positive real numbers p_j satisfying

$$\sum_{j \in I} p_j = 1,$$

such that

$$\rho = \sum_{j \in I} p_j \rho_j^A \otimes \rho_j^B.$$

Otherwise, ρ is called **entangled**.

Quantum Entanglement : A review

For a pure state $|\psi\rangle \in \mathbb{H}^A \otimes \mathbb{H}^B$, view this as a mixed state by writing

$$\rho = |\psi\rangle\langle\psi|.$$

Then we see that it is separable if and only if

$$|\psi\rangle = |\phi\rangle \otimes |\theta\rangle, \quad \text{for some } |\phi\rangle \in \mathbb{H}^A \quad \text{and} \quad |\theta\rangle \in \mathbb{H}^B.$$

Quantum Entanglement : A review

Example

A Bell state $|\Phi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ is an entangled state. Indeed, they are **maximally entangled**.

Example

A GHZ state $|GHZ\rangle = \frac{|0\rangle^{\otimes M} + |1\rangle^{\otimes M}}{\sqrt{2}}$.

Quantum Entanglement : A review

Definition

A pure state $|\Psi\rangle$ in the tensor product of identical Hilbert spaces $\mathbb{H}^A$ is said to be **maximally entangled** if

$$\rho^A(\Psi) = \lambda 1, \quad \text{with } 0 < \lambda < 1.$$

Meanwhile, for an entangled state, i.e an element in $\mathcal{H}_1 \otimes \mathcal{H}_2$, we want to reduce the system into one Hilbert space $\mathcal{H}_1$. To this end, the **partial trace** is defined by the unique linear operator

$$\text{Tr}_{\mathcal{H}_2} : L(\mathcal{H}_1 \otimes \mathcal{H}_2) \rightarrow L(\mathcal{H}_1), \quad \text{Tr}_{\mathcal{H}_2}(R \otimes S) = \text{Tr}(S)R.$$

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No-Cloning Theorem : A review

Definition

Given (1) an arbitrary (original) state $|\psi\rangle \in \mathbb{H}$ to be copied and (2) a (white-page) state $|\omega\rangle \in \mathbb{H}$ to emerge as a copy, a **quantum-copier** K is a linear transformation that leaves the original state $|\psi\rangle$ unchanged and transforms $|\omega\rangle$ such that it becomes $|\psi\rangle$, i.e., K is an operator satisfying

$$K : |\psi\rangle \otimes |\omega\rangle \mapsto |\psi\rangle \otimes |\psi\rangle$$

for arbitrary $|\psi\rangle \in \mathbb{H}$ and a given fixed $|\omega\rangle \in \mathbb{H}$.

Theorem (Quantum No-Cloning Theorem)

A quantum-copier cannot exist.

Proof of No-Cloning Theorem

We must have

$$K(|0\rangle \otimes |\omega\rangle) = |0\rangle \otimes |0\rangle, \quad K(|1\rangle \otimes |\omega\rangle) = |1\rangle \otimes |1\rangle,$$

and also,

$$K\left(\frac{|1\rangle + |0\rangle}{\sqrt{2}} \otimes |\omega\rangle\right) = \frac{|1\rangle + |0\rangle}{\sqrt{2}} \otimes \frac{|1\rangle + |0\rangle}{\sqrt{2}}.$$

However, we also have

$$K\left(\frac{|1\rangle + |0\rangle}{\sqrt{2}} \otimes |\omega\rangle\right) = \frac{1}{\sqrt{2}}(|1\rangle \otimes |1\rangle + |0\rangle \otimes |0\rangle).$$

Quantum Teleportation

Dense Quantum Coding : Transmit two classical bits by only sending one qubit

Quantum Teleportation : Transmit one qubit by sending two classical bits.

Basic Theme : Use entanglement, in particular use Bell state.

Bell states and measurement : A review

Notation :

1. Bell basis

$$\begin{aligned} |\Phi^\pm\rangle &:= \frac{1}{\sqrt{2}} \left(|00\rangle \pm |11\rangle \right) \\ |\Psi^\pm\rangle &:= \frac{1}{\sqrt{2}} \left(|01\rangle \pm |10\rangle \right). \end{aligned}$$

2. Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

3. By direct computation, one has

$$\begin{aligned}(\sigma_z \otimes \sigma_z)|\Phi^\pm\rangle &= |\Phi^\pm\rangle \\(\sigma_z \otimes \sigma_z)|\Psi^\pm\rangle &= -|\Psi^\pm\rangle \\(\sigma_x \otimes \sigma_x)|\Phi^\pm\rangle &= \pm|\Phi^\pm\rangle \\(\sigma_x \otimes \sigma_x)|\Psi^\pm\rangle &= \pm|\Psi^\pm\rangle.\end{aligned}$$

Therefore, we can distinguish each Bell basis vectors by measuring with respect to the operators $\sigma_z \otimes \sigma_z$ and $\sigma_x \otimes \sigma_x$.

We start with **dense quantum coding**.

Alice wants to send two classical bits x_1, x_0 to Bob.

1. Prepare a Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.
2. Apply a quantum gate $U^A(x_1, x_0)$ to $|\Phi^+\rangle$.
3. Measures $\sigma_z^A \otimes \sigma_z^B$ and $\sigma_x^A \otimes \sigma_x^B$ and observes two values (one of 1, -1 each).

Dense quantum coding

Table 6.1 Protocol of dense quantum coding

Alice wants to send the classical bits x_1x_0	So she applies $U^A(x_1x_0)$	The state of the total system becomes $(U^A \otimes \mathbf{1}) \Phi^+\rangle$	on which Bob measures $\sigma_z^A \otimes \sigma_z^B$ and $\sigma_x^A \otimes \sigma_x^B$ and observes the values
00	$\mathbf{1}^A$	$ \Phi^+\rangle$	+1 , +1
01	σ_z^A	$ \Phi^-\rangle$	+1 , -1
10	σ_x^A	$ \Psi^+\rangle$	-1 , +1
11	$\sigma_z^A \sigma_x^A$	$ \Psi^-\rangle$	-1 , -1

$$\begin{aligned}
 U^A(00) &= \mathbf{1}^A \\
 U^A(01) &= \sigma_z^A \\
 U^A(10) &= \sigma_x^A \\
 U^A(11) &= \sigma_z^A \sigma_x^A .
 \end{aligned}
 \tag{6.6}$$

Quantum Teleportation : Theory

1. Alice wants to teleport the qubit $|\psi\rangle = a|0\rangle + b|1\rangle \in \mathbb{H}^S$ to Bob.
2. For an entangled Bell state,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|0\rangle^A \otimes |0\rangle^B + |1\rangle^A \otimes |1\rangle^B) \in \mathbb{H}^A \otimes \mathbb{H}^B,$$

suppose part of it belongs to Alice and Bob has the remainder.

3. Alice then applies $\sigma_z^S \otimes \sigma_z^A$ and $\sigma_x^S \otimes \sigma_x^A$ to her qubit in $\mathbb{H}^S \otimes \mathbb{H}^A$.
4. According to the result of measurement, the three-qubit total state is determined.
5. Bob determines the operator to apply and get the qubit $|\psi\rangle$.

Quantum Teleportation : Theory

Important computation

$$\begin{aligned} |\psi\rangle^S \otimes |\Psi^+\rangle^{AB} &= \left(a|0\rangle^S + b|1\rangle^S \right) \otimes \frac{1}{\sqrt{2}} \left(|0\rangle^A \otimes |0\rangle^B + |1\rangle^A \otimes |1\rangle^B \right) \\ &= \frac{1}{2} \left[|\Psi^+\rangle^{SA} \otimes |\psi\rangle^B + |\Psi^+\rangle^{SA} \otimes \left(\sigma_x^B |\psi\rangle^B \right) \right. \\ &\quad \left. + |\Psi^-\rangle^{SA} \otimes \left(\sigma_z^B |\psi\rangle^B \right) + |\Psi^-\rangle^{SA} \otimes \left(\sigma_x^B \sigma_z^B |\psi\rangle^B \right) \right], \end{aligned}$$

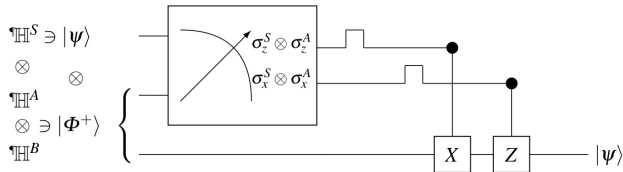
where we used

$$|0\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}} \left(|\Psi^+\rangle^{SA} + |\Psi^-\rangle^{SA} \right), \dots$$

Quantum Teleportation : Theory

Table 6.2 Protocol for teleportation

Alice measures $\sigma_z^S \otimes \sigma_z^A$ and $\sigma_x^S \otimes \sigma_x^A$ to observe	The three-qubit total state after measurement is then: $ \Psi\rangle^{SA} \otimes$ Bob's qubit-state	From bits received Bob determines to apply U^B	The state of Bob's qubit then becomes
+1 , +1	$ \Phi^+\rangle \otimes \psi\rangle$	$\mathbf{1}^B$	$(\mathbf{1}^B)^2 \psi\rangle = \psi\rangle$
+1 , -1	$ \Phi^-\rangle \otimes \sigma_z^B \psi\rangle$	σ_z^B	$(\sigma_z^B)^2 \psi\rangle = \psi\rangle$
-1 , +1	$ \Psi^+\rangle \otimes \sigma_x^B \psi\rangle$	σ_x^B	$(\sigma_x^B)^2 \psi\rangle = \psi\rangle$
-1 , -1	$ \Psi^-\rangle \otimes \sigma_x^B \sigma_z^B \psi\rangle$	$\sigma_z^B \sigma_x^B$	$\sigma_z^B \sigma_x^B \sigma_x^B \sigma_z^B \psi\rangle = \psi\rangle$



Quantum Teleportation : Implementation

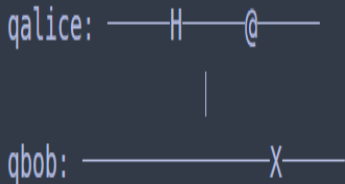
```
import cirq
import cirq_google
import numpy as np

import random
# Create qubits : msg = qubit[0], galice = qubit[1], qbob = qubit[2]
qubit=[0] *(3)
qubit[0] = cirq.NamedQubit('msg')
qubit[1] = cirq.NamedQubit('galice')
qubit[2] = cirq.NamedQubit('qbob')
print(qubit)
circuit = cirq.Circuit()

[cirq.NamedQubit('msg'), cirq.NamedQubit('galice'), cirq.NamedQubit('qbob')]
```

Quantum Teleportation : Implementation

```
#Create a Bell state entangled pair  
circuit.append([cirq.H(qubit[1]), cirq.CNOT(qubit[1],qubit[2])])  
print(circuit)
```



Quantum Teleportation : Implementation

```
#Create a random state for the message , repeating this may cause error
ranX = random.random()
ranY = random.random()
print(ranX, ranY)
circuit.append([cirq.X(qubit[0]) ** ranX, cirq.Y(qubit[0]) ** ranY])
print(circuit)
```

0.7511826460115856 0.7631230865776707

msg: ————X^{0.751}———Y^{0.763}———

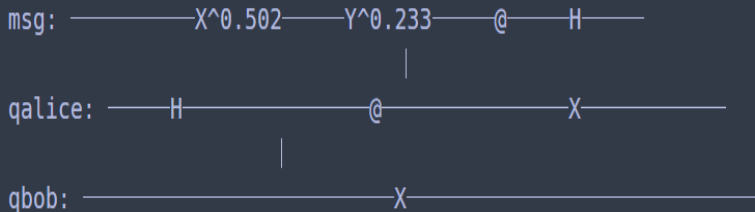
qalice: ———H———@———

|

qbob: ————X———

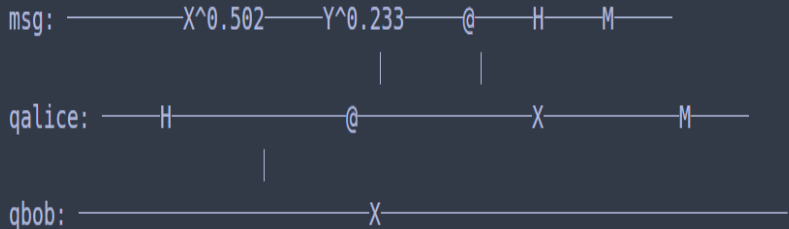
Quantum Teleportation : Implementation

```
#Unitary operator rotating the two-qubit basis of the message  
#and Alice's entangled qubit  
circuit.append([cirq.CNOT(qubit[0],qubit[1]), cirq.H(qubit[0])])  
print(circuit)
```



Quantum Teleportation : Implementation

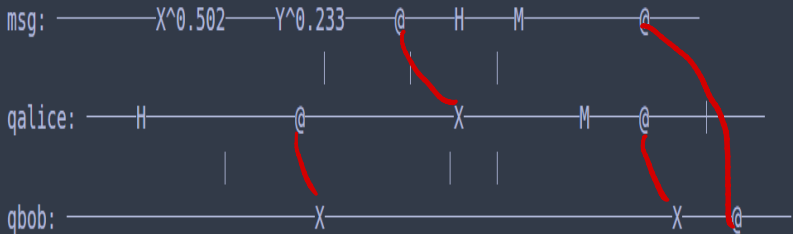
```
#Measurement  
circuit.append(cirq.measure(qubit[0], qubit[1]))  
print(circuit)
```



Quantum Teleportation : Implementation

#Using two classical bits to recover the original message

```
circuit.append([cirq.CNOT(qubit[1],qubit[2]), cirq.CZ(qubit[0], qubit[2])])  
print(circuit)
```



Quantum Teleportation : Implementation

```
sim = cirq.Simulator()
q0 = cirq.LineQubit(0)
message = sim.simulate(cirq.Circuit([cirq.X(q0)**ranX, cirq.Y(q0)**ranY]))
print("\nBloch Sphere of the Message qubit in the initial state:")
expected = cirq.qis.bloch_vector_from_state_vector(message.final_state_vector, 0)
print(" x: ", np.around(expected[0], 4), " y:", np.around(expected[1],4), " z: ", np.around(expected[2],4))
```

Bloch Sphere of the Message qubit in the initial state:

x: -0.0036 y: -1.0 z: -0.004



Quantum Teleportation : Implementation

```
final_results = sim.simulate(circuit)
print("\nBloch Sphere of Bob's qubit in the final state:")
teleported = cirq.bloch_vector_from_state_vector(final_results.final_state_vector, 2)
print("x: ", np.around(teleported[0], 4), " y: ", np.around(teleported[1], 4), " z: ", np.around(teleported[2], 4))
print("\nBloch Sphere of the Message qubit in the final state:")
message_final = cirq.bloch_vector_from_state_vector(final_results.final_state_vector, 0)
print("x: ", np.around(message_final[0], 4), " y: ", np.around(message_final[1], 4), " z: ", np.around(message_final[2], 4))
```

Bloch Sphere of Bob's qubit in the final state:

x: -0.0036 y: -1.0 z: -0.004



Bloch Sphere of the Message qubit in the final state:

x: 0.0 y: 0.0 z: 1.0

Some comments on Quantum Teleportation

Interesting paper : Pirandola, S.; Eisert, J.; Weedbrook, C.; Furusawa, A.; Braunstein, S. L. (2015). "[Advances in Quantum Teleportation](#)".
Nature Photonics

1. What we saw in previous slide is the qubit-teleportation ($d = 2$):
Input is pure, entangled state is a Bell pair, and we apply a qubit Pauli operator.
2. For continuous-variable system ($d = \infty$) : The systems are bosonic modes, input is a coherent state, entangled state is an EPR state, and operator is a phase-space displacement.
3. Entanglement swapping : "Entanglement" itself can be teleported, i.e. the qubit to be transferred can be part of an entangled state.
4. Quantum gates teleportation : One can teleport quantum gates. Experimentally, two-qubit gates have been teleported.

Some comments on Quantum Teleportation

1. Issue of Fidelity : Teleportation fidelity should exceed an appropriate threshold.

Given to density operators ρ and σ , the fidelity $F(\rho, \sigma)$ is defined by

$$F(\rho, \sigma) = \left(\text{Tr}(\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}) \right)^2,$$

for the case of pure states, we have $F(\rho, \sigma) = |\langle \psi_\rho, \psi_\sigma \rangle|^2$.

For example, we have $F(\rho, \sigma) = F(\sigma, \rho)$ and $F \in [0, 1]$.

2. A continuous variable teleportation system cannot reach a 100% fidelity.

THANK YOU FOR YOUR LISTENING